

Spatial heterogeneity of carbon dioxide, methane, and ammonia in a naturally ventilated dairy barn: high-resolution mapping and implications for sampling design

Harsh Sahu^{a,b}, Long Chen^a, Thomas Amon^{a,b}, David Janke^{a,*}

^aLeibniz Institute for Agricultural Engineering and Bioeconomy (ATB), Max-Eyth-Allee 100, 14469 Potsdam, Germany

^bInstitute of Animal Hygiene and Environmental Health, Department of Veterinary Medicine, Freie Universität Berlin, Germany

*Corresponding author: djanke@atb-potsdam.de

Abstract

Homogeneous gas mixing is commonly assumed when concentrations are used to characterise air quality or calculate emissions from naturally ventilated dairy barns, but the magnitude and spatial structure of departures from this assumption remain poorly resolved. Carbon dioxide (CO₂), methane (CH₄), and ammonia (NH₃) concentrations were therefore mapped at 51 internal locations arranged as 17 vertical triplets. Two cavity ring-down spectrometers and two Fourier-transform infrared spectrometers were operated at a temporal resolution of four minutes; the Fourier-transform infrared measurements were harmonised to the cavity ring-down spectroscopy scale using a preliminary 24 h co-located comparison. All gases exhibited structured vertical and horizontal heterogeneity. Mean concentrations decreased from top to middle to bottom, but the magnitude of the vertical gradient differed among gases and horizontal positions. The CH₄/CO₂ and NH₃/CO₂ ratios also decreased with height group from top to bottom, showing that the observed structure was not explained solely by uniform dilution or concentration. As a practical consequence of this spatial analysis, alternative height configurations were evaluated in 743 complete two-hour blocks. Top-only and bottom-only measurements produced systematic, gas-specific biases, whereas a spatially distributed top-and-bottom network reproduced the full-density mean with Lin's concordance correlation coefficients of 0.999 and mean relative biases below 0.4%. The primary finding is that gas concentrations within the barn were spatially structured rather than homogeneous; understanding this structure subsequently enabled the network to be reduced from 51 to 34 locations without materially changing the barn mean.

Keywords: air quality; dairy housing; horizontal gradient; mixing ratio; spatial variability; vertical profile

Nomenclature

Symbol	Definition
CH ₄	methane
CO ₂	carbon dioxide
CCC	Lin's concordance correlation coefficient
CRDS	cavity ring-down spectroscopy
FTIR	Fourier-transform infrared spectroscopy
H ₂ O	water vapour
NH ₃	ammonia
NVDB	naturally ventilated dairy barn
RE	relative deviation from the full-density estimate, %
RMSE	root mean square error
<i>s</i>	south-background sampling location
SE	standard error
SP	sampling point

1 Introduction

Emission estimates for naturally ventilated livestock buildings depend on both the air exchange rate and the concentration difference across the measurement boundary. Neither component is readily constrained because wind and buoyancy can reverse flow through large openings, while animals, manure surfaces, internal obstructions, and mechanical mixing devices generate local sources and transport pathways. Uncertainty in a concentration measurement is therefore not only analytical; it also includes whether the selected location represents the air volume or control surface of interest. These distinctions are important because a representative internal concentration is not necessarily representative of an emission flux when the corresponding airflow field is unknown (Calvet et al., 2013; Ogink et al., 2013).

The consequences of imperfect mixing have been demonstrated under controlled and field conditions. In a mechanically ventilated test installation, ventilation-rate errors of up to 86% arose when tracer concentrations from individual internal positions were used, although this value describes a tracer-gas ventilation estimate rather than a barn concentration or emission error (Van Buggenhout et al., 2009). Wind-tunnel measurements of a large naturally ventilated building likewise showed that emission estimates based on concentration and velocity at one constant height could differ markedly from vertically

resolved measurements (Janke et al., 2020). These studies establish that the spatial arrangement of measurements can be as important as their nominal precision, but their controlled flow fields do not reproduce the full temporal variability of an occupied dairy barn.

Field studies have identified heterogeneity in both horizontal and vertical directions. Height-dependent differences in NH_3 , CO_2 , and tracer-gas mixing ratios were observed in a naturally ventilated dairy cow barn, with more consistent mixing above the animal-occupied zone (Mendes et al., 2015). Gas-specific vertical profiles and incomplete correlations among heights were also reported in an open-sided dairy barn (D’Urso et al., 2024). Horizontally, the mean and its variability were sensitive to the number and separation of sampling locations (D’Urso et al., 2022). External wind speed and direction further altered internal concentrations and air exchange in a site-specific manner (Saha et al., 2013). Consequently, a vertical profile observed at one horizontal position cannot be assumed to apply throughout a barn, nor can the behaviour of one gas be transferred directly to another.

Spatial representativeness must also be tested across contrasting temporal regimes. In naturally ventilated buildings, the pressure distribution around the envelope changes with wind speed and approach direction, whereas the relative contribution of buoyancy changes with the indoor–outdoor temperature difference. The resulting ventilation regime can alter both the magnitude and the spatial ordering of sampling-point concentrations (Saha et al., 2013). Measurements made during one meteorological period therefore cannot establish whether a spatial pattern is structurally persistent or conditional on the prevailing weather. Repeated dense-network observations in summer and autumn, followed by a reduced-network application in late autumn and winter, provide a means of examining that persistence. Such comparisons require concurrent meteorological information and careful separation of season from campaign, analyser deployment, and sampling design; regional observations can describe the synoptic setting but cannot reproduce the barn-scale airflow field.

Gas ratios provide an additional view of mixing because responses shared by two gases can partly cancel, whereas differences in source position, reactivity, or transport remain. Spatial variability in ammonia-to-tracer mixing ratios has shown that apparent agreement at one height can coexist with horizontal differences elsewhere in the building (Mendes et al., 2015). Simulation-based work has consequently investigated representative sampling heights (Doubria et al., 2021), and information-based approaches have been used to identify non-redundant sensor locations across simulated ventilation cases (Chen et al., 2026). However, the latter approach was developed for CFD-derived local mean age of air and related flow variables; its direct transfer to sequential field measurements of gas concentration has not been established.

Dense field sampling can characterise these patterns but increases tubing, switching-

cycle duration, maintenance requirements, and the risk of sampling line failure. Conversely, reduction based only on sensor count or a geometric centre may discard vertical extremes or persistent local effects. Sampling density and spatial arrangement have both affected airflow estimates in a naturally ventilated mock-up building (De Voeleer et al., 2017), yet evidence is limited on how a full-scale, three-dimensional concentration network can be reduced while retaining both its dense-network mean and meaningful spatial structure over extended field campaigns.

Accordingly, three questions were addressed. First, how large and persistent is the spatial heterogeneity of CO_2 , CH_4 , NH_3 , and their ratios across height, horizontal position, and temporal aggregation? It was hypothesised that location and height effects would be gas-specific and would persist most strongly for NH_3 and nitrogen-containing ratios. Second, can the middle locations be omitted while retaining both the dense-network concentration estimate and the observed vertical structure? It was hypothesised that a spatially distributed top-and-bottom design would agree more closely with the 51-location reference than either extreme alone while preserving vertical contrast. Third, can one physically eligible location provide a pragmatic approximation of the network mean? It was hypothesised that a stable candidate could be ranked, but that gas- and campaign-specific residual bias would preclude universal reference status.

2 Materials and methods

2.1 Experimental barn

Measurements were conducted in the experimental dairy barn of the Lehr- und Versuchsanstalt für Tierzucht und Tierhaltung, Gross Kreutz, Brandenburg, Germany. The naturally ventilated barn was 38.88 m long and 17.65 m wide. The asymmetrical fibre-cement roof reached approximately 6.2 m at the ridge and approximately 3.6 m at the sidewalls. The internal airspace was approximately 4529 m³. The loose-housing barn was designed for 58 dairy cows and contained littered cubicles, concrete walkways, a manure scraper, and adjacent milking and young-stock areas.

2.2 Sampling layout

Fifty-one internal SPs were distributed over 17 horizontal positions (Fig. 1). Three vertically aligned locations were installed at each position: a top location 60 cm below the roof, a middle location 360 cm above the floor, and a bottom location 260 cm above the floor. The location numbers were assigned consecutively within each triplet. Thus, locations 1, 4, ..., 49 represented the top group; locations 2, 5, ..., 50 represented the middle group; and locations 3, 6, ..., 51 represented the bottom group. A single south-background SP,

denoted s , was treated conceptually as location 52 but was excluded from all analyses of internal spatial heterogeneity.

2.3 Gas analysers and preliminary field comparison

Concentrations of CO_2 , CH_4 , NH_3 , and water vapour were measured using two CRDS analysers (Picarro G2508; Picarro Inc., Santa Clara, CA, USA; CRDS8 and CRDS9) and two FTIR analysers (Gasmet CX4000; Gasmet Technologies Oy, Vantaa, Finland; FTIR1 and FTIR2). Before the first mapping campaign, all four analysers were operated for 24 h at a co-located central barn position. Two polytetrafluoroethylene sampling tubes were used: one was divided with a T-piece between the two CRDS analysers and the other was divided between the two FTIR analysers. The FTIR instruments measured CO_2 and CH_4 concentrations that were 6% higher than those measured by CRDS, while FTIR NH_3 was 9% higher. Accordingly, FTIR measurements from Campaign 1 were harmonised to the CRDS scale as

$$C_{\text{CO}_2, \text{corr}} = C_{\text{CO}_2, \text{raw}} / 1.06, \quad (1)$$

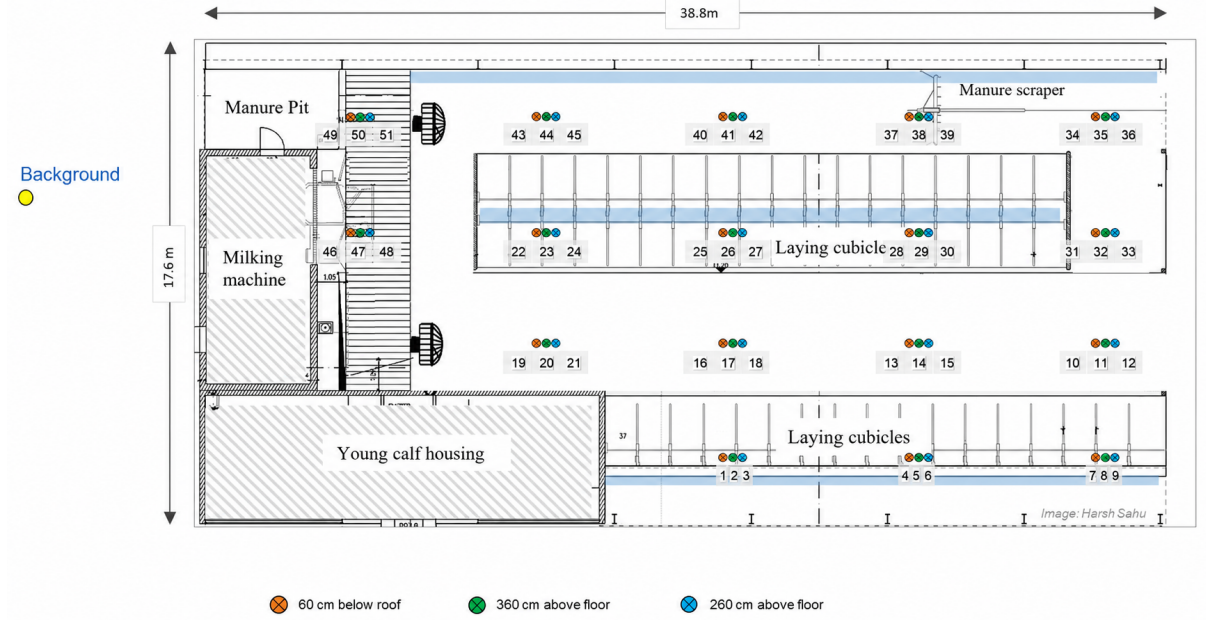
$$C_{\text{CH}_4, \text{corr}} = C_{\text{CH}_4, \text{raw}} / 1.06, \quad (2)$$

$$C_{\text{NH}_3, \text{corr}} = C_{\text{NH}_3, \text{raw}} / 1.09. \quad (3)$$

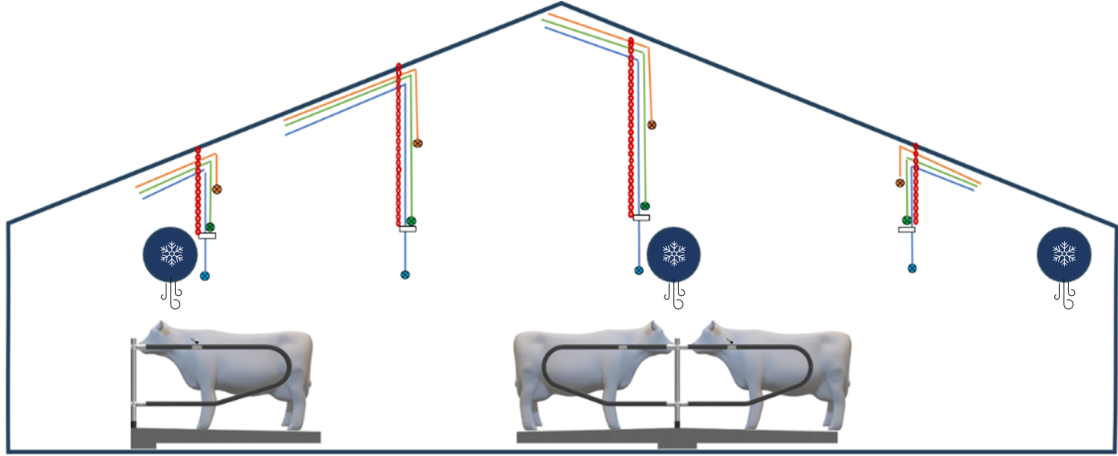
CRDS measurements and all water-vapour measurements were retained without correction. Both raw and corrected concentrations were preserved in the analytical dataset. Before and after the field comparison, all analysers were tested with certified cylinders containing approximately 700 ppm CO_2 , 7 ppm CH_4 , and 5 ppm NH_3 ; deviations remained within the manufacturer-stated $\pm 5\%$ accuracy. The field comparison was therefore used for cross-platform alignment under representative sampling conditions. The CRDS instruments were treated as the operational comparison scale, not as an absolute metrological reference. Consequently, the correction removed the observed mean platform offset but could not separate instrument-specific drift from spatial effects during the subsequent fixed analyser assignments.

2.4 Measurement campaigns and data processing

Campaign 1 data were recovered for June–August 2024. Concurrent ordered cycling by all four analysers was identified from 3 June to 16 July and again from 1 to 26 August, with shorter interruptions retained as gaps. CRDS9 sampled locations 1–16, FTIR2 sampled locations 17–26, CRDS8 sampled locations 27–42, and FTIR1 sampled locations 43–51 and s . The resulting network comprised 51 internal locations and the south-background location. Gas measurements were recorded at a four-minute temporal resolution. Each



(a) Plan view.



(b) Cross-sectional representation of the sampling heights and fans.

Figure 1: Experimental barn and high-resolution sampling design. Orange symbols indicate locations 60 cm below the roof, green symbols indicate locations 360 cm above the floor, and blue symbols indicate locations 260 cm above the floor. The numbered internal locations comprise 17 vertical triplets. The yellow point labelled Background represents the south-background location s , which was retained in the dataset but excluded from internal spatial comparisons.

settled CRDS valve step was identified from the ordered 1–16 sequence. The transition signal was excluded, the first 60 s was discarded, and the subsequent 180 s was averaged. Across June–August, CRDS9 contained 22,384 valid four-minute steps and two isolated three-minute steps; CRDS8 contained 26,099 four-minute steps and one three-minute step. For every three-minute step, 120 s remained after the common 60 s flushing period. Single-position and irregular sequences were retained in audit tables but excluded from the campaign data. FTIR exports were cleaned by removing repeated embedded headers and retaining the required timestamp, sampling-position, and gas fields. Their instrument timestamps were retained without an additional flushing operation. All timestamps were parsed in the Europe/Berlin time zone, duplicate combinations of timestamp, analyser, and location were removed within each campaign, and the retained rows were ordered by timestamp, analyser, and location.

Campaign 2 was conducted from 16 November 2024 at 00:06:47 to 31 December 2024 at 23:57:10 using CRDS8 and CRDS9. The 17 middle locations were omitted following the Campaign 1 density assessment. Locations 19 and 40 were also excluded because of fan proximity and a sampling-line leakage problem, respectively, resulting in 32 internal locations. Each CRDS sampling step lasted 240 s; the first 60 s was discarded for flushing and the remaining 180 s was averaged.

Non-positive gas records were treated as missing in the manuscript analytical dataset. Carbon dioxide values below 300 ppm were also treated as invalid transition or analyser records because they were physically implausible for barn or ambient air. No row was deleted from the immutable source dataset.

Table 1: Measurement design and analytical records retained for the two campaigns. The south-background location *s* was retained in the data but excluded from the count of internal locations.

Campaign Period		Analysers	Vertical design	Internal SPs	Rows
1	Jun.–Aug. 2024	CRDS8, CRDS9, FTIR1, FTIR2	top, middle, bottom	51	111 210
2	16 Nov.–31 Dec. 2024	CRDS8, CRDS9	top and bottom	32	20 632

2.5 Regional meteorological observations

Hourly regional meteorological observations were obtained from the Deutscher Wetterdienst Climate Data Center (DWD CDC) for Potsdam station 03987, located approximately 18.5 km from the barn (Deutscher Wetterdienst Climate Data Center, 2024). The station coordinates were 52.3812° N and 13.0622° E, and its elevation was 81 m. Historical hourly wind, air-temperature and moisture, and precipitation products were used. DWD timestamps were parsed as UTC and converted to Europe/Berlin local time. The retained

variables were wind speed, meteorological wind direction, air temperature, relative humidity, and hourly precipitation.

Observations were classified according to the experimental periods: Campaign 1 summer (1 June–31 August 2024), Campaign 1 autumn (1–24 October 2024), and Campaign 2 late autumn/winter (16 November–31 December 2024). Wind roses were constructed from 16 directional sectors and six wind-speed classes. Daily mean temperature and relative humidity, the daily temperature range, and daily precipitation total were calculated for temporal visualisation. These regional data were used descriptively in the present manuscript version; they were not included in the concentration models reported below. Where available, the on-site 10 m mast remains the preferred source for subsequent modelling of barn-specific wind effects.

2.6 Gas ratios

The relative mixing ratios of CH_4 and NH_3 to CO_2 were calculated for positive, valid concentrations:

$$R_{\text{CH}_4:\text{CO}_2} = 100 C_{\text{CH}_4}/C_{\text{CO}_2}, \quad (4)$$

$$R_{\text{NH}_3:\text{CO}_2} = 100 C_{\text{NH}_3}/C_{\text{CO}_2}, \quad (5)$$

$$R_{\text{NH}_3:\text{CH}_4} = 100 C_{\text{NH}_3}/C_{\text{CH}_4}. \quad (6)$$

The ratios are dimensionless and are reported as percentages. Where inferential modelling of ratios is subsequently performed, their logarithmic forms, $\log(C_{\text{CH}_4}/C_{\text{CO}_2})$ and $\log(C_{\text{NH}_3}/C_{\text{CO}_2})$, are preferred to avoid the asymmetry of untransformed ratios.

2.7 Evaluation of sampling-density reduction

Daily means at each location were first calculated to provide independent blocks for descriptive means and 95% confidence intervals. For direct comparison of sampling configurations, measurements were averaged into two-hour blocks. A block was retained only when all 51 internal locations were represented, yielding 743 complete blocks in the recovered dataset.

Seven vertical configurations were reconstructed from the Campaign 1 data: top only, middle only, bottom only, top plus middle, middle plus bottom, top plus bottom, and the full three-height design. The full 51-location mean was used as the reference estimate within each block. For a reduced configuration k , the relative deviation was calculated as

$$RE_{k,t} = 100 \frac{\bar{C}_{k,t} - \bar{C}_{51,t}}{\bar{C}_{51,t}}. \quad (7)$$

The top-and-bottom configuration was selected a priori as the operational reduced design because it retained both ends of the observed vertical gradient. Its agreement with the full design was quantified using mean bias, relative bias, RMSE, coefficient of determination, and Lin’s CCC. Spatial heterogeneity within each block was expressed as the standard deviation of log-transformed location concentrations.

Following Chen et al. (2026), the temporal informativeness of every location was described using Shannon entropy. For each gas and ratio, location means from complete two-hour blocks were divided into four equal-width bins. The empirical probabilities p_b were used to calculate

$$H^* = -\frac{\sum_{b=1}^4 p_b \log_2(p_b)}{\log_2(4)}. \quad (8)$$

This was an empirical adaptation of the cross-case CFD entropy method: the present cases were observed two-hour blocks rather than simulated ventilation scenarios. Sensitivity to temporal aggregation was assessed at 2, 4, 8, and 24 h resolutions.

2.8 Multiscale persistence and reference-location analysis

To determine whether location differences diminished through time, location means were calculated at 1, 2, 4, 8, 12, 24, 72, and 168 h resolutions. Within each contemporaneous block, every location was compared with the spatial mean of the measured network. The decay in median absolute relative error was described by a power-law regression of log error against log averaging duration. Daily location-rank stability was quantified using pairwise Spearman correlations and Kendall’s coefficient of concordance, W .

In addition, descriptive statistics were calculated explicitly at hourly, daily, weekly, and whole-campaign scales for all three concentrations and the three ratios. For each location and scale these comprised observation count, mean, median, minimum, 5th and 95th percentiles, maximum, standard deviation (SD), median absolute deviation (MAD), and coefficient of variation (CV). Whole-campaign statistics were calculated directly from all valid observations rather than by averaging weekly summaries. Repeated-location Friedman tests were used to assess whether SD, MAD, and CV differed among the four reporting scales.

Variance was partitioned using linear mixed-effects models of log-transformed concentrations and ratios, with random intercepts for location and date and fixed effects for campaign, height, and hour. Campaign-specific generalised additive models included cyclic hour-of-day smooths, longer-term date smooths, and location random effects. PCA and correlation-distance hierarchical clustering were applied to hourly location relative-error profiles. Cluster number was selected by the greatest mean silhouette width.

Reference-location selection was defined before statistical ranking. Location 40 was excluded because of confirmed sampling-line leakage and location 19 because of local fan influence. A conservative physical sensitivity screen also excluded the fan corridors

containing locations 19–21 and 43–45. Only locations measured in both campaigns were eligible. Within each gas and campaign, candidates were ranked for absolute bias, MAE, RMSE, temporal stability, Spearman association and four-bin mutual information with the contemporaneous spatial mean. Equally weighted percentile scores were then averaged across CO₂, CH₄, and NH₃; ratios were retained as a sensitivity analysis.

Predictive pattern recognition compared a linear model, cyclic GAM, and regression tree for hourly location relative error. Five-fold cross-validation was grouped by complete date, preventing observations from the same date from appearing in both training and test data. Change-point screening of daily barn means and spatial CV was exploratory and was not interpreted causally without corresponding operational records.

All cleaning, calculations, and figures were produced in R version 4.5.3 (R Foundation for Statistical Computing, Vienna, Austria). Data manipulation was conducted using `data.table`; mixed-effects and additive models were fitted using `lme4` and `mgcv`, respectively; clustering and silhouette analysis used `cluster`; regression trees used `rpart`; and figures were produced using `ggplot2`. The complete analysis code and intermediate summary tables were retained to permit the sampling configurations and ranking criteria to be reproduced. Top, middle, and bottom locations were displayed in orange, green, and light blue, respectively.

3 Results

3.1 High-resolution concentration patterns

Marked spatial and vertical variation was detected during Campaign 1 (Fig. 2). Across internal locations, the mean corrected CO₂ concentrations were approximately 616 ppm at the top, 611 ppm at the middle, and 609 ppm at the bottom. Corresponding CH₄ means were 18.84, 18.51, and 18.38 ppm, while NH₃ means were 2.05, 2.04, and 2.02 ppm. The direction and magnitude of the vertical gradient were not uniform among horizontal positions. Particularly distinct values occurred at locations 19 and 40, supporting their exclusion from the subsequent operational design.

3.2 Effect of vertical sampling configuration

Use of one height alone produced systematic deviations from the full-density mean (Fig. 3; Table 2). Top-only sampling overestimated CO₂, CH₄, and NH₃ by mean values of 0.68%, 1.01%, and 1.18%, respectively. Bottom-only sampling underestimated the same gases by 0.50%, 0.78%, and 1.40%. The middle-only configuration was closer to the full mean, but removed the vertical extremes required for evaluating heterogeneity.

Combining top and bottom locations substantially reduced configuration bias. The mean absolute relative errors were 0.33% for CO₂, 0.95% for CH₄, and 0.77% for NH₃.

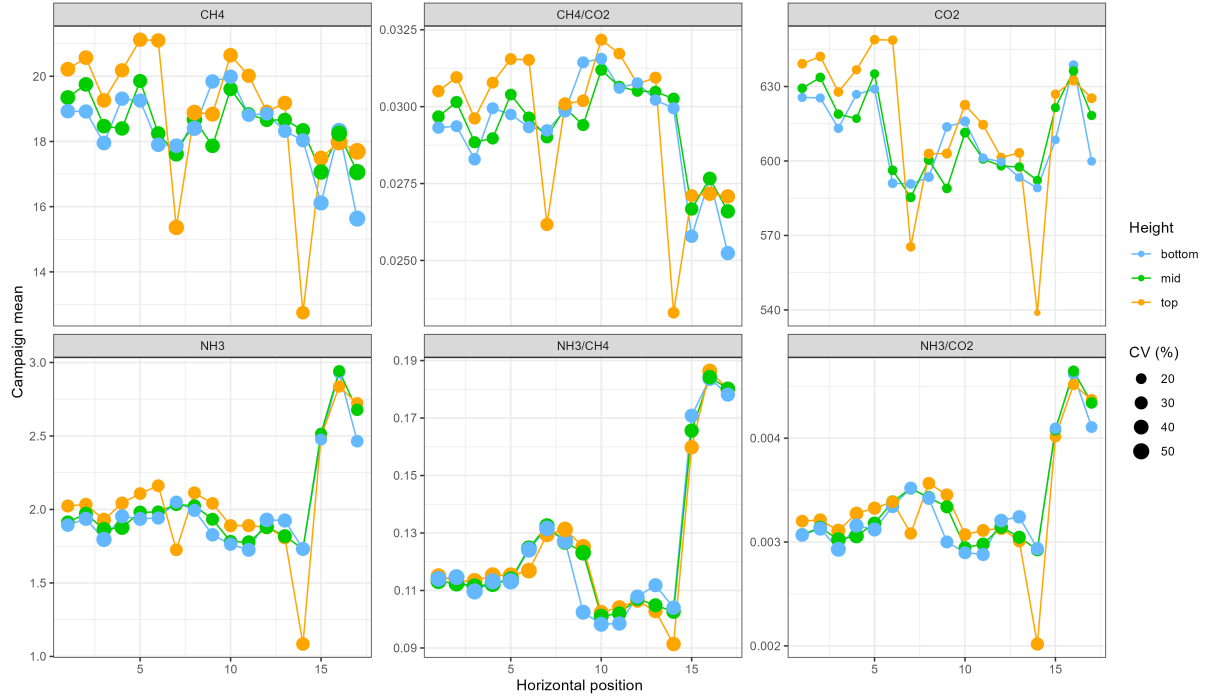


Figure 2: Campaign 1 corrected concentration and gas-ratio means across the 17 horizontal positions. Symbol size represents the location-specific temporal coefficient of variation, capped at 100% for display. Orange, green, and light-blue symbols denote top, middle, and bottom locations, respectively. The south-background location s was excluded.

286 Their 95th percentiles were 0.85, 2.52, and 1.92%, respectively.

287 3.3 Entropy and temporal-resolution sensitivity

288 The most temporally informative location depended on the response (Fig. 4). The highest
 289 normalised entropy occurred at locations 4, 7, and 15 for CO_2 , CH_4 , and NH_3 , respectively;
 290 no single location was universally informative. Increasing the averaging interval from 2
 291 to 24 h reduced the median spatial CV from 7.73 to 4.60% for CO_2 , from 19.11 to 9.56%
 292 for CH_4 , and from 22.88 to 20.11% for NH_3 . The persistent ammonia CV indicated that
 293 its heterogeneity was not primarily a short-timescale artefact.

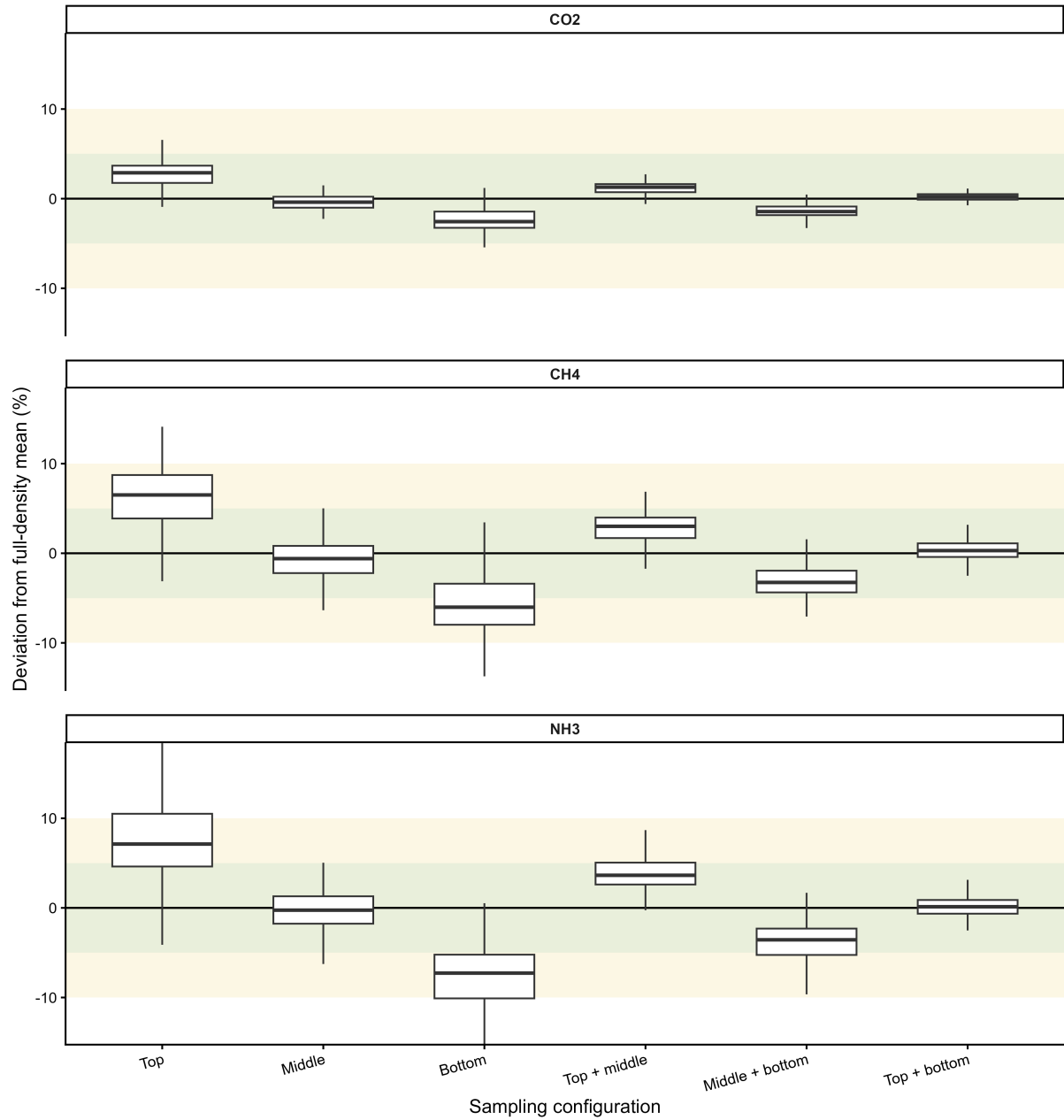


Figure 3: Deviation of six reduced vertical sampling configurations from the full 51-location mean in 743 complete two-hour blocks. Boxes show the interquartile range and centre lines show medians; whiskers extend to 1.5 times the interquartile range. Green and yellow bands indicate deviations within $\pm 5\%$ and $\pm 10\%$, respectively. Configurations labelled top, middle, or bottom contain 17 locations; paired-height configurations contain 34 locations. Outliers are omitted visually but retained in all summaries.

Table 2: Performance of selected reduced configurations relative to the full 51-location mean. Values are calculated across 743 complete two-hour blocks. MAD denotes median absolute deviation and P_{95} denotes the 95th percentile of absolute deviation.

Gas	Configuration	Mean RE (%)	MAD (%)	P_{95} (%)
CO ₂	Top only	0.68	0.82	3.71
	Middle only	-0.18	0.52	1.70
	Bottom only	-0.50	0.78	3.28
	Top + bottom	0.09	0.26	0.85
CH ₄	Top only	1.01	2.42	9.75
	Middle only	-0.23	1.46	5.03
	Bottom only	-0.78	2.17	8.37
	Top + bottom	0.12	0.73	2.52
NH ₃	Top only	1.18	2.34	9.67
	Middle only	0.22	1.28	3.84
	Bottom only	-1.40	2.02	8.04
	Top + bottom	-0.11	0.64	1.92

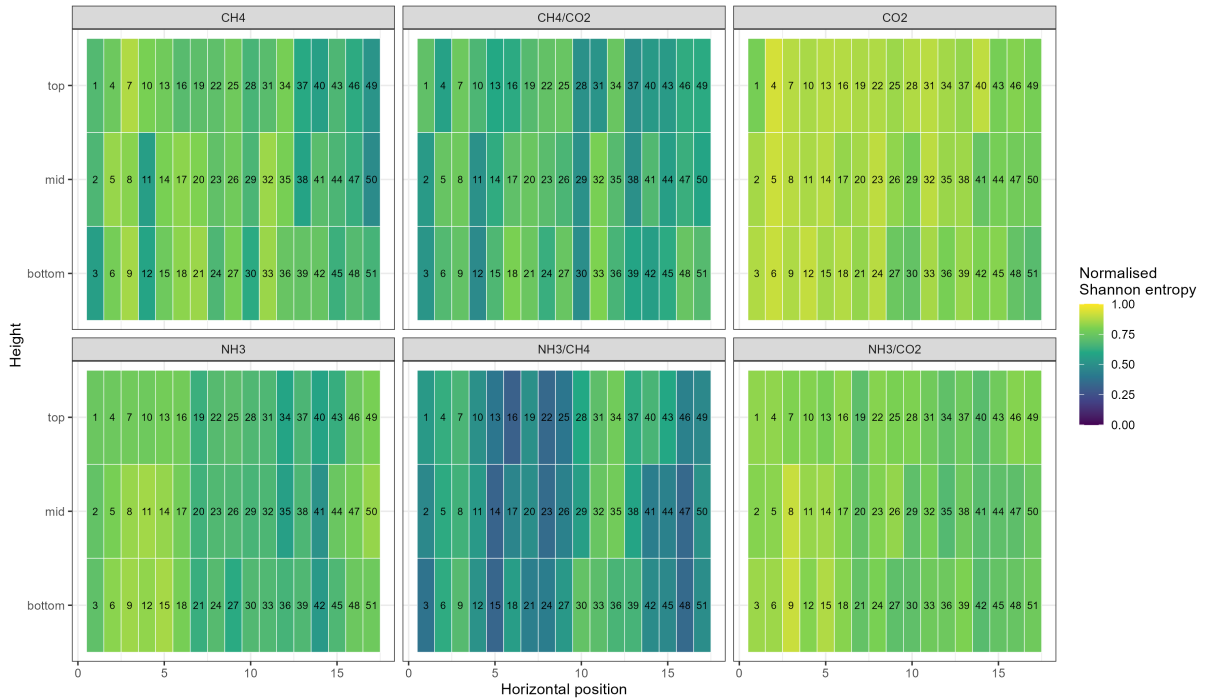


Figure 4: Normalised four-bin Shannon entropy of corrected gas concentrations and gas ratios at each Campaign 1 location. Entropy was calculated across 743 complete two-hour blocks and normalised by $\log_2(4)$. Numbers identify sampling locations. Higher values indicate broader and more even occupation of the four empirical bins, not necessarily a greater mean concentration.

3.4 Concentrations and variation over time

Barn-wide daily means varied substantially within both analysed periods. Across the 89 represented Campaign 1 dates, the daily mean (mean across dates) was 607.6 ppm for CO₂, 18.14 ppm for CH₄, and 2.08 ppm for NH₃. Their daily means ranged from 506.0 to 684.3 ppm, from 9.87 to 22.92 ppm, and from 1.25 to 3.12 ppm, respectively. The between-date CV was 6.36% for CO₂, 14.31% for CH₄, and 16.89% for NH₃. Thus, ammonia showed the largest relative day-to-day variation during the dense summer period.

Across the 47 represented Campaign 2 dates, the corresponding daily means were 636.9 ppm, 18.40 ppm, and 0.91 ppm. Daily means ranged from 490.1 to 802.7 ppm for CO₂, from 7.33 to 31.24 ppm for CH₄, and from 0.53 to 1.64 ppm for NH₃. Between-date CVs increased to 13.31%, 36.16%, and 27.83%, respectively. Carbon dioxide and methane reached their lowest and highest daily barn means on the same dates within each campaign, whereas the ammonia extrema occurred on different dates. This temporal decoupling was consistent with the gas-specific spatial and ratio behaviour. These summaries describe the June–August component of Campaign 1 and Campaign 2; the recovered October component will require the same calculation before an autumn comparison is reported.

3.5 Regional meteorological conditions during the study periods

The regional meteorological setting differed visibly among the three study periods (Fig. 5). Westerly winds were most frequent during the summer period, with a secondary easterly component. The October period had a broader directional distribution, including westerly, south-easterly, and northerly sectors. During Campaign 2, winds from west and south-west dominated and the higher wind-speed classes occurred more frequently. Mean hourly wind speed increased from 3.60 m s⁻¹ in summer to 4.06 m s⁻¹ in October and 4.83 m s⁻¹ during Campaign 2.

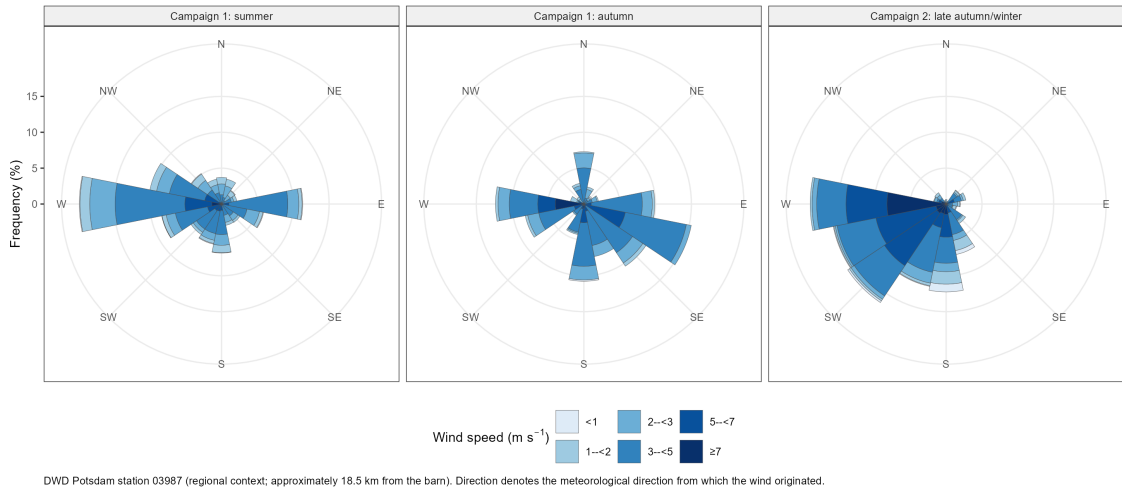


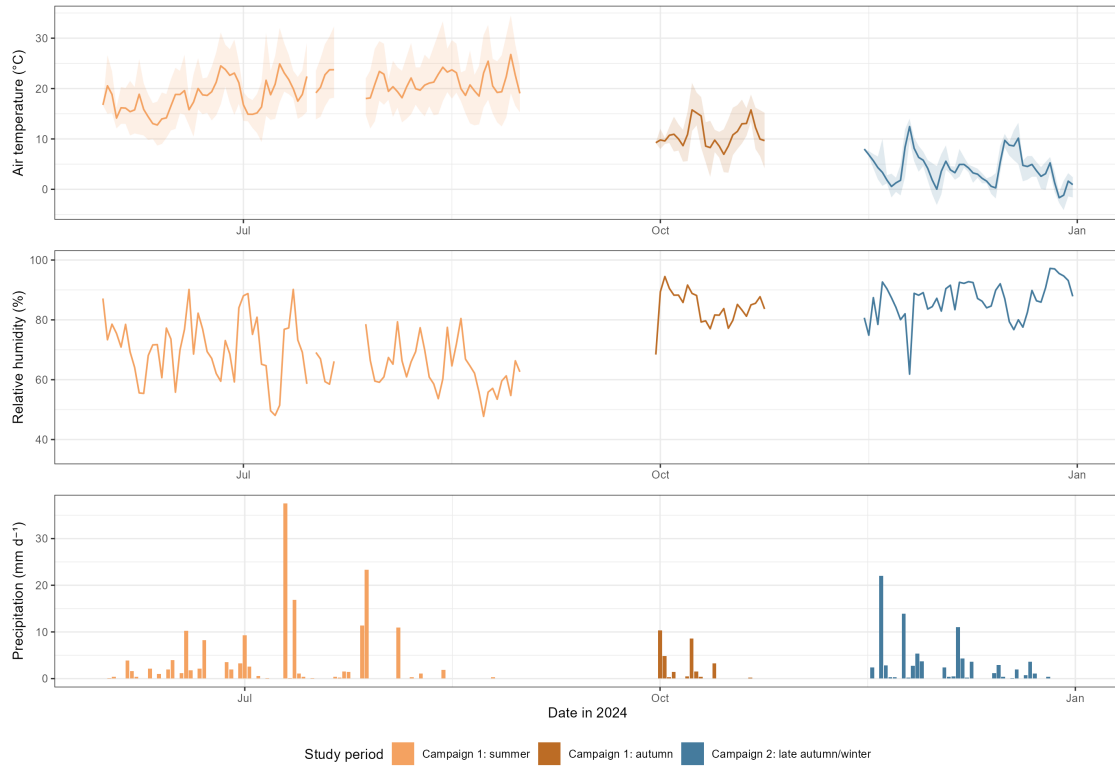
Figure 5: Wind roses for the three experimental periods constructed from hourly DWD Potsdam station 03987 observations. Colours show wind-speed classes and radial length shows the percentage of valid hourly observations in each directional sector. Direction denotes the meteorological direction from which the wind originated. The station is approximately 18.5 km from the barn; the figure therefore describes regional conditions rather than barn-local airflow.

Air temperature declined and relative humidity increased across the study periods (Fig. 6). Mean temperature was 19.72 °C in summer, 10.95 °C in October, and 4.07 °C during Campaign 2. Corresponding mean relative humidities were 67.4%, 84.8%, and 86.6%. Total precipitation within the respective study windows was 169.2 mm, 31.3 mm, and 88.5 mm; these totals are not normalised for the unequal durations of the periods. The contrasts establish that the gas measurements spanned distinct regional weather regimes, but do not by themselves demonstrate meteorological effects on concentration.

3.6 Persistence, convergence, and pattern recognition

Spatial differences became smaller but did not disappear with longer averaging (Fig. 7). In Campaign 1, median absolute location error decreased from 6.43 to 3.12% for CO₂, from 15.65 to 4.86% for CH₄, and from 17.15 to 12.01% for NH₃ between hourly and weekly aggregation. Campaign 2 errors decreased from 5.73 to 3.65%, 17.91 to 10.24%, and 17.34 to 15.37%, respectively. All power-law exponents were negative, although convergence was weakest for NH₃.

The explicit four-scale summaries supported this distinction (Fig. 8). Median within-location CV decreased from hourly to weekly resolution in both campaigns: from 16.33 to 4.40% and from 16.38 to 6.39% for CO₂; from 38.37 to 9.45% and from 48.17 to 19.79% for CH₄; and from 31.47 to 18.32% and from 28.60 to 13.25% for NH₃ in Campaigns 1 and 2, respectively. The same general attenuation occurred for all three ratios. Friedman tests showed scale effects on SD, MAD, and CV for every concentration and ratio in both campaigns (all $P < 10^{-12}$). In contrast, whole-campaign CV calculated from all raw



Daily summaries from DWD Potsdam station 03987. Shaded temperature bands show the daily minimum–maximum range. Missing intervals between study periods are intentionally retained.

Figure 6: Daily regional air temperature, relative humidity, and precipitation during the experimental periods from DWD Potsdam station 03987. Lines show daily means; the temperature ribbon shows the daily minimum–maximum range; bars show daily precipitation totals. Gaps between experimental periods are retained intentionally. These observations provide regional meteorological context and are not measurements of the barn microclimate.

observations remained high because it retained the complete seasonal and diurnal range; it is therefore a descriptor of total campaign variability, not an additional temporal-averaging step.

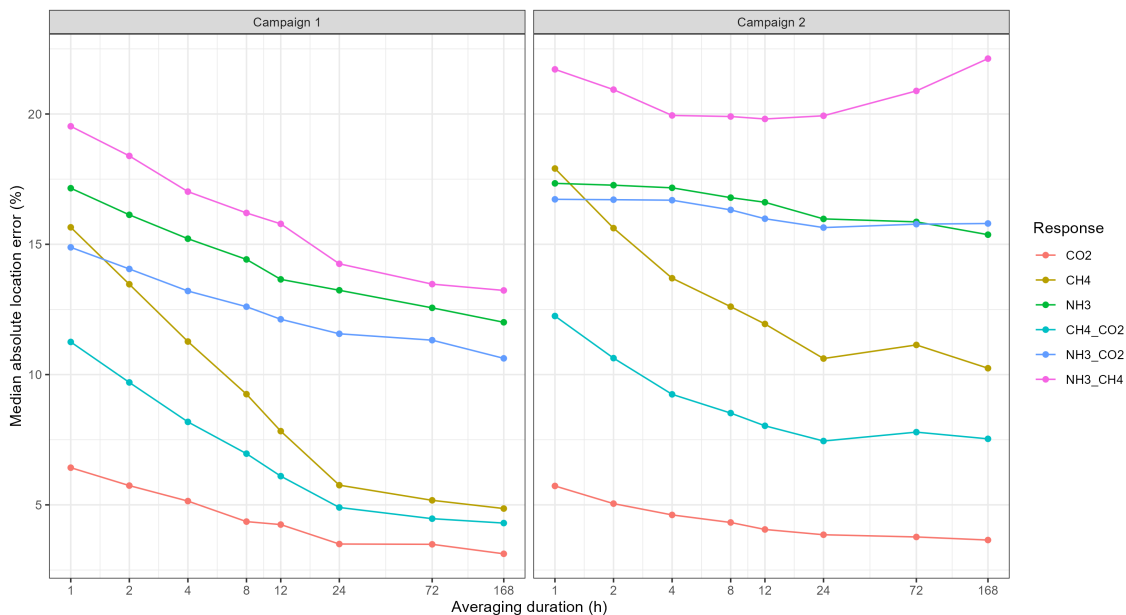


Figure 7: Convergence of median absolute location error as the averaging duration increased from 1 to 168 h. Errors were calculated relative to the contemporaneous measured-network mean. The logarithmic horizontal axis shows that concentration fields became more representative with aggregation but retained gas- and campaign-specific residual heterogeneity.

Daily rank stability was moderate in Campaign 1: Kendall’s W was 0.433 for CO_2 , 0.447 for CH_4 , and 0.523 for NH_3 . Location accounted for 4.4% of log- CO_2 variance, 3.7% of log- CH_4 variance, and 21.2% of log- NH_3 variance. The corresponding location fractions were 4.9% for CH_4/CO_2 , 22.9% for NH_3/CO_2 , and 20.6% for NH_3/CH_4 . PCA and clustering did not identify one universal grouping for all gases, indicating response-specific spatial patterns.

Regression trees outperformed linear and GAM benchmarks in date-grouped cross-validation, but predictive performance remained moderate. Mean test R^2 was 0.205 for CO_2 , 0.121 for CH_4 , and 0.409 for NH_3 . Location and clock time therefore captured only part of hourly spatial error, leaving substantial variation for ventilation, weather, animal, and operational covariates.

3.7 Pragmatic reference-location selection

Location 25 ranked first among physically eligible cross-campaign candidates (Fig. 9). It was outside the conservative fan corridors, was measured in both campaigns, and remained first when ratio behaviour was added as a sensitivity criterion. Its hourly biases in Campaigns 1 and 2 were -1.07 and +0.62% for CO_2 , +3.00 and +0.88% for CH_4 , and

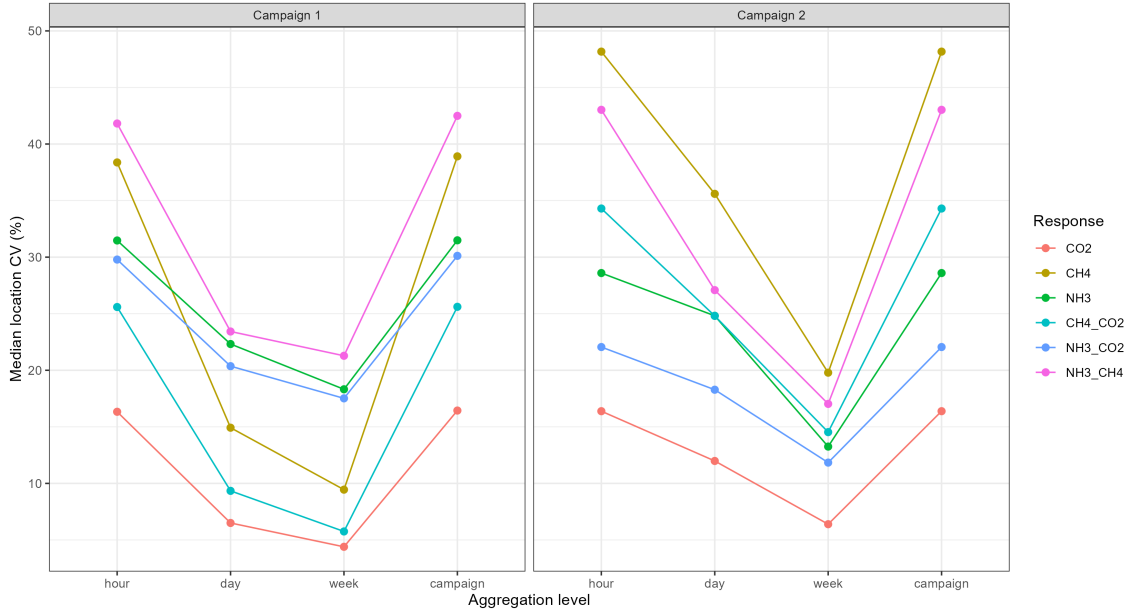


Figure 8: Median within-location coefficient of variation (CV) for CO₂, CH₄, NH₃, and their three ratios at hourly, daily, weekly, and whole-campaign reporting scales. Hourly, daily, and weekly values quantify variability among block means, whereas the whole-campaign value was calculated directly from every valid observation and therefore preserves the full seasonal and diurnal range. Consequently, the whole-campaign bar should not be interpreted as a continuation of the temporal averaging sequence.

-2.09 and +7.25% for NH₃, respectively. It was therefore the best single-point compromise rather than a universally unbiased substitute for spatial averaging.

3.8 Agreement between full and reduced designs

The top-and-bottom mean closely reproduced the full 51-location estimate throughout Campaign 1 (Fig. 11). Lin's CCC was 0.999 for all three gases. Mean relative biases were 0.09% for CO₂, 0.12% for CH₄, and -0.11% for NH₃. The corresponding RMSE values were 2.69 ppm, 0.23 ppm, and 0.019 ppm. Agreement for the spatial heterogeneity metric was lower than agreement for the mean but remained strong, with CCC values between 0.953 and 0.968.

3.9 Longer-term application of the reduced design

The reduced-height principle was applied from mid-November to the end of December 2024 in Campaign 2 (Fig. 12). Mean concentrations remained higher at top than at bottom locations: CO₂ averaged approximately 640 and 625 ppm, CH₄ averaged 18.6 and 17.4 ppm, and NH₃ averaged 0.96 and 0.83 ppm at the top and bottom, respectively. The mean CH₄/CO₂ ratios were 2.75% and 2.64%, and the mean NH₃/CO₂ ratios were 0.149% and 0.131% at the top and bottom.

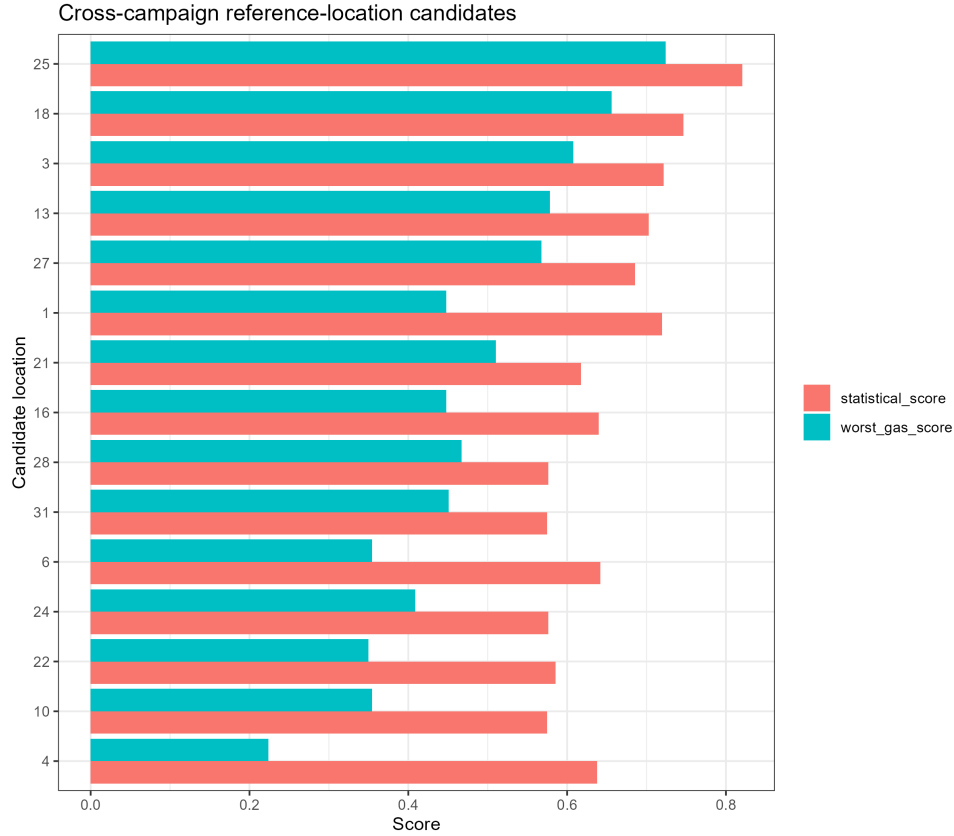


Figure 9: Leading cross-campaign reference-location candidates. The primary score equally combines concentration bias, absolute error, temporal stability, correlation, and mutual information after the physical exclusion criteria. The worst-gas score penalises candidates whose performance was poor for any one gas. Fan-corridor candidates were retained only in the sensitivity ranking.

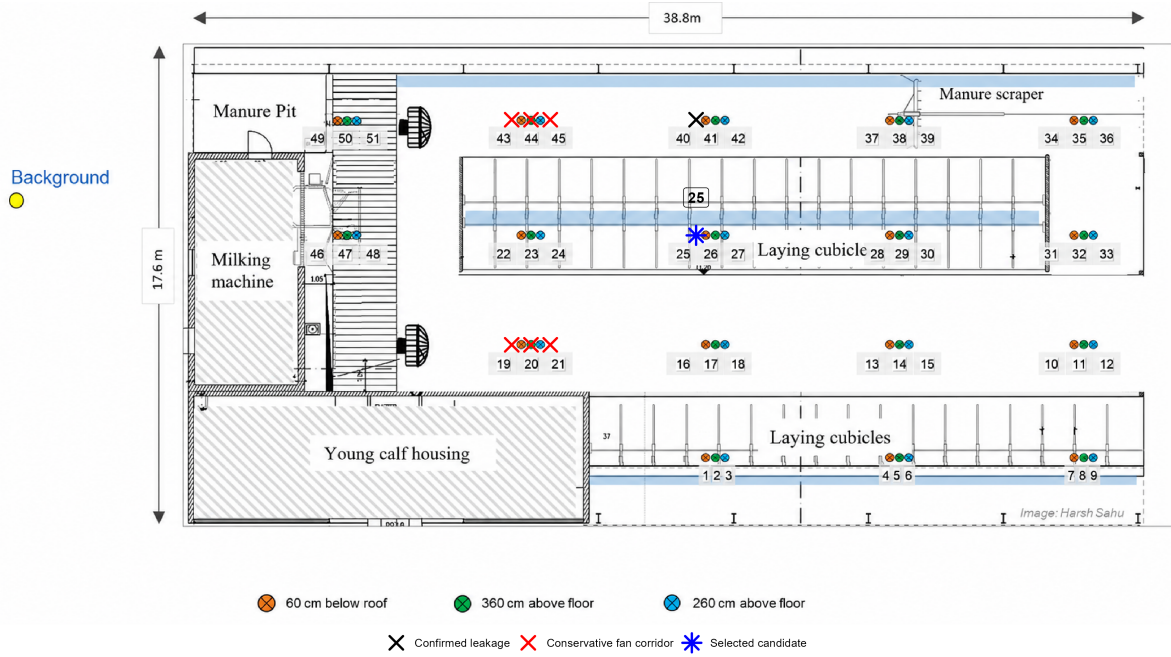


Figure 10: Physical screening of the statistically selected reference candidate. Location 25 is marked in blue. Red crosses identify the conservatively excluded fan corridors containing locations 19–21 and 43–45; the black cross marks location 40, which was excluded because of confirmed sampling-line leakage. The exact fan axes should be confirmed against the operational installation before final engineering acceptance.

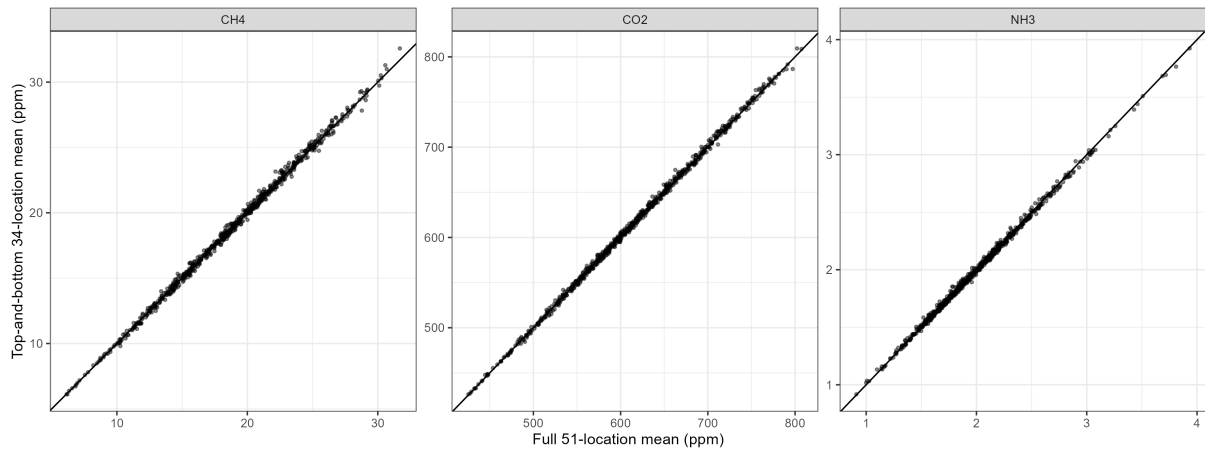


Figure 11: Agreement between the full 51-location mean and the reduced 34-location top-and-bottom mean in 743 complete two-hour blocks. Solid lines represent exact 1:1 agreement. CCC denotes Lin's concordance correlation coefficient, and bias is the mean relative difference from the full-density estimate.

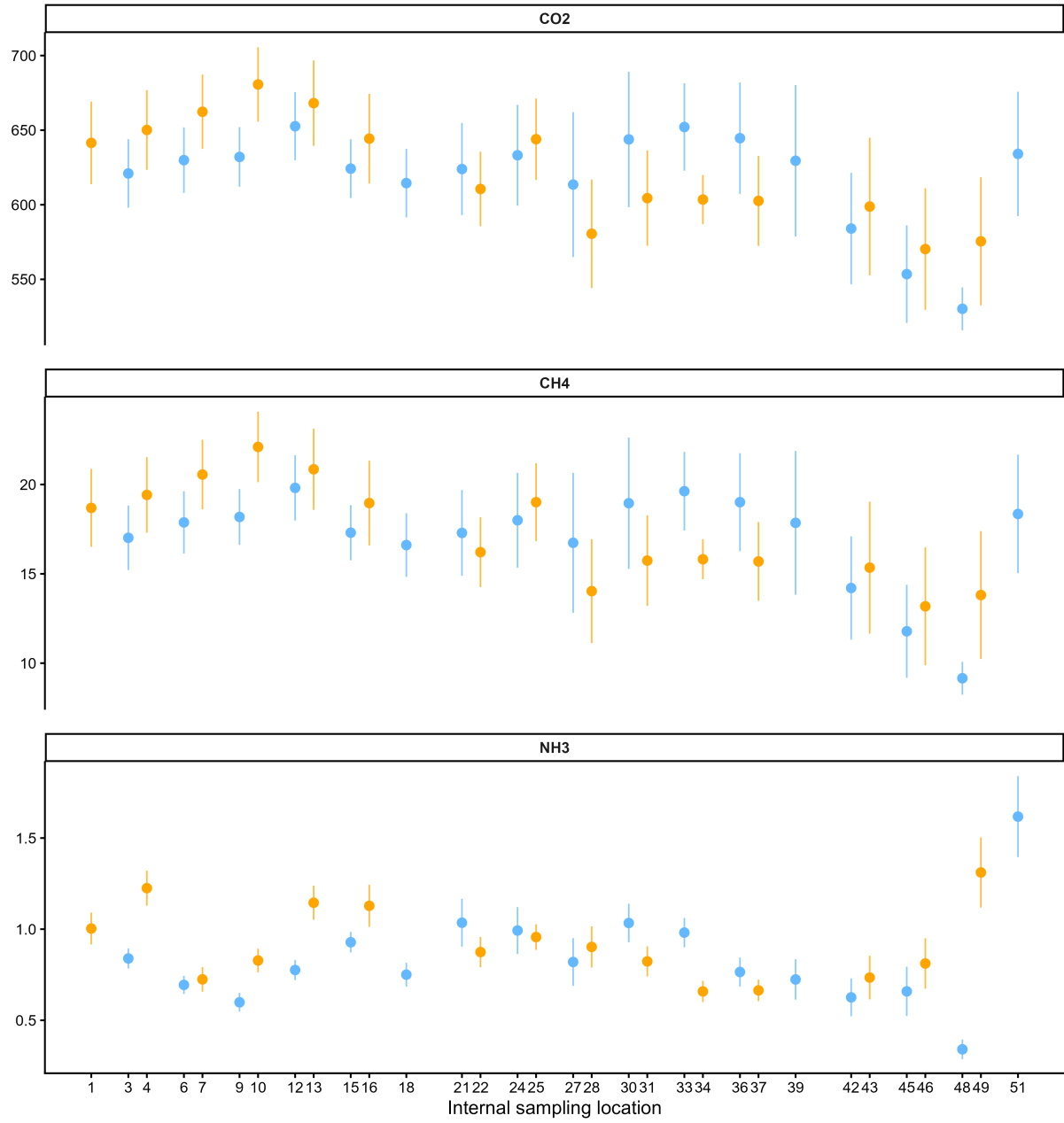


Figure 12: Campaign 2 concentrations measured with the reduced top-and-bottom network. Points denote means of daily location means and vertical bars denote 95% confidence intervals calculated across daily blocks. Orange and light-blue symbols denote top and bottom locations, respectively.

4 Discussion

4.1 Gas-specific spatial structure

The measurements did not support homogeneous concentration fields within the investigated barn. Vertical differences were systematic at the network level, but their magnitude varied among gases and horizontal positions. This gas-specific structure is consistent with earlier field observations showing height-dependent NH_3 , CO_2 , and tracer-gas mixing ratios (Mendes et al., 2015), and with the incomplete agreement among three measurement heights reported for an open dairy barn (D’Urso et al., 2024). The present study extends that evidence by resolving the vertical comparison at 17 horizontal positions rather than at one or a few profiles.

Horizontal coverage remained important even when the vertical design was reduced. This agrees with D’Urso et al. (2022), for whom both the number and spacing of horizontal locations affected the estimated concentration distribution. The current results further show why vertical and horizontal design cannot be considered independently: omission of one height may be acceptable for the spatial mean only when the retained heights remain distributed across the barn. The variance partitioning also demonstrated that this requirement was response-specific. Location explained a substantially larger fraction of variation in NH_3 and its ratios than in CO_2 or CH_4 , indicating that a network adequate for one gas cannot automatically be assumed adequate for another.

Several mechanisms are compatible with the observed patterns, although they were not measured directly in these campaigns. Carbon dioxide was dominated by a spatially distributed animal source, whereas methane included localised enteric release and ammonia was linked to manure-contaminated surfaces and gas-surface exchange. Wind direction, opening configuration, buoyancy, and fan operation could then transport or dilute these sources differently. External wind conditions have previously affected internal concentrations and air exchange in a naturally ventilated dairy building (Saha et al., 2013), but the absence of concurrent airflow-vector measurements prevents individual features in the present concentration maps from being attributed causally to a specific opening, fan, or flow path.

4.2 Information contained in the gas ratios

The ratios were not interpreted as source signatures in isolation. Instead, they were used as diagnostics of whether gases varied proportionally across space and time. The comparatively small location fraction for CH_4/CO_2 suggests that part of the spatial variation in these two animal-associated gases was shared. In contrast, the larger location fractions for NH_3/CO_2 and NH_3/CH_4 indicate that ammonia retained spatial behaviour not removed by normalisation to either carbon gas. This finding is consistent with the spatial variabil-

ity of ammonia-to-tracer mixing ratios reported by Mendes et al. (2015) and reinforces the need to evaluate ratio stability rather than assume co-dispersion.

Ratio interpretation remains conditional on the source distributions and background treatment. No background-corrected enhancement regression was applied, and the south-background location was excluded from the internal heterogeneity analysis. Consequently, the ratios quantify proportionality of the measured internal concentrations; they do not by themselves distinguish animal, manure, or ventilation contributions and should not be interpreted as emission factors.

4.3 Temporal aggregation does not replace spatial coverage

4.3.1 Temporal concentration dynamics

The daily barn means show that temporal variability was not merely scatter among sampling locations. Concentrations changed coherently at the barn scale, but the relative magnitude and timing differed among gases. The coincident daily extrema of CO_2 and CH_4 are compatible with shared effects of animal occupancy, activity, and ventilation, while the different timing and larger relative variation of NH_3 are compatible with additional control by manure-surface conditions and gas-surface exchange. These are supported interpretations rather than causal tests because animal activity, surface state, airflow, and meteorology have not yet been included together in the fitted models.

Campaign 2 had wider daily relative variation for all three gases, but its absolute NH_3 level was lower than in the analysed summer part of Campaign 1. This difference cannot be assigned to season alone: campaign, network density, omitted locations, analyser composition, ventilation regime, and unmodelled barn operation changed together. The recovered October dense period offers the more defensible first test of seasonal persistence because it retains the Campaign 1 spatial design. Its integration with on-site mast and regional meteorological data should determine whether changes in temporal concentration level coincide with changes in spatial ranks or only with barn-wide dilution.

The DWD figures confirm that the measurement periods differed in regional wind direction, wind speed, temperature, humidity, and rainfall. They do not yet establish which of those variables influenced the temporal concentration variation. In particular, the lower regional temperature and higher wind speed in Campaign 2 coincided with wider daily variation in CO_2 and CH_4 , but campaign design and barn operation changed at the same time. The regional observations should therefore be treated as candidate explanatory variables rather than post-hoc causes. Formal evaluation requires hourly alignment with gas-concentration blocks, circular treatment of wind direction, non-linear weather terms, date-level grouping, and comparison with the on-site mast observations.

Aggregation reduced median absolute location error for all three gases, but the rate and remaining error differed. The stronger persistence of NH_3 relative to CO_2 and CH_4

was consistent with its larger location variance fraction. Thus, short-lived fluctuations were averaged out while stable or repeatedly occurring spatial contrasts remained. The result clarifies two different meanings of representativeness: a longer average can improve the precision of a location mean without removing its systematic displacement from the contemporaneous network mean.

Whole-campaign variability should also not be read as the final point of a monotonic averaging sequence. The campaign CV was calculated from all valid observations and therefore retained diurnal and seasonal variation, whereas the daily and weekly CVs described variation among aggregated block means. The high whole-campaign CV therefore reflects a broader temporal domain, not a failure of weekly averaging. Sequential sampling adds a further limitation: rapid changes during a switching cycle can be aliased as spatial differences, and adsorption or delayed response is particularly relevant for ammonia (Rom and Zhang, 2010). Complete two-hour blocks reduced, but could not eliminate, this temporal-spatial ambiguity.

4.4 Reduction of the vertical network

The middle-only design reproduced the dense-network mean comparatively well, but it suppressed the vertical extremes that defined the heterogeneity under investigation. The top-and-bottom design retained these extremes and allowed their opposing deviations to balance in the spatial mean. Its near-unity CCC and small bias demonstrate agreement with the 51-location reference estimate for the observed two-hour blocks; they do not prove that every local plume or measure of spatial spread was preserved.

The finding is consistent with the broader observation that sensor placement is not interchangeable with sensor number. Different arrangements at the same sampling density produced different airflow estimates in a naturally ventilated mock-up building (De Voogeleer et al., 2017). Similarly, vertically composite concentration and velocity measurements were more representative than a single-height approach in a wind-tunnel study (Janke et al., 2020). Those studies concern airflow or emission estimation, whereas the present validation concerns internal concentration means. The agreement across these different outcomes supports vertically distributed sampling as a design principle, but does not make their numerical errors directly comparable.

Campaign 2 demonstrated that a top-and-bottom network could be operated over a longer late-autumn and winter period. It did not provide an independent 51-versus-34 validation because no simultaneous dense reference was present and two site-specific locations had also been removed. The evidence therefore supports a reduction from 51 to 34 locations within Campaign 1 and an operational 32-location application in Campaign 2, rather than a universal 32-location optimum.

4.5 Information-based ranking and the single-point question

The entropy analysis adapted the information-based placement principle described by Chen et al. (2026). In that study, sensor informativeness was evaluated across CFD scenarios using local mean age of air and related flow variables. Here, entropy described temporal diversity of observed gas concentrations at individual locations. It was therefore a diagnostic component rather than a direct implementation of the CFD-informed optimisation algorithm, and the reduced vertical design had been defined from the configuration comparison rather than selected by entropy alone.

Likewise, the grouped-date predictive models showed only moderate out-of-date performance. This is scientifically informative: location, height, and clock time captured repeatable structure, but much of the hourly relative error remained unexplained. Weather, fan status, animal distribution, and barn microclimate are plausible omitted predictors. The models should therefore be viewed as pattern-recognition benchmarks, not operational concentration predictors.

The recovered autumn dense-network period is particularly important for the next model revision because it retains the same 51-location geometry and four-analyser deployment used in the summer period. It can therefore test whether location ranks, entropy, and reduced-design agreement persist under a different meteorological regime without simultaneously changing the spatial network. Wind speed should be allowed a non-linear response, and direction should be represented by sine and cosine components or pre-specified wind sectors rather than as a linear 0–360° covariate. Temperature and relative humidity are physically justified covariates, while precipitation is more appropriately evaluated as an event indicator and lagged modifier than assumed to exert an immediate independent effect. These analyses have not yet been used to revise the numerical results reported above.

Location 25 was the best compromise among candidates that passed the predefined physical screen and were available in both campaigns. Its small carbon-gas biases support its use as a pragmatic single-point approximation within this barn. However, the NH_3 bias changed to +7.25% in Campaign 2, demonstrating residual response- and campaign-dependence. A point that approximates the internal concentration mean is also not automatically suitable for emission calculation because flux depends jointly on concentration and airflow across a defined control surface (Calvet et al., 2013; Ogink et al., 2013). Location 25 should therefore be treated as a testable site-specific candidate rather than a reference standard.

4.6 Limitations and future research

The principal limitations define the scope of inference. The summer and autumn periods belong to Campaign 1 and retain the dense design, whereas the late-autumn and winter

period belongs to Campaign 2 and uses a reduced network and different analyser deployment. Consequently, differences between Campaigns 1 and 2 cannot be attributed causally to season alone. Comparisons between the summer and autumn parts of Campaign 1 provide a less confounded test of temporal persistence, but analyser and location remained partly confounded because the four analysers were assigned to fixed spatial ranges. The applied factors corrected mean FTIR–CRDS offsets but did not establish traceability to an absolute reference or remove subsequent drift. Sequential switching prevented an instantaneous three-dimensional snapshot, and the 51-location mean remained a dense-network reference estimate rather than a volume-weighted true concentration.

Future validation should randomise or exchange analyser assignments. Sampling line response should be quantified for the deployed tube lengths and flow rates, and reduced and dense networks should be compared concurrently in independent periods. Meteorological, fan, animal-distribution, and barn-climate variables should be aligned with the concentration blocks to test the proposed mechanisms. Wind measured at the on-site 10 m mast should be treated as the primary external-flow observation where available. Hourly DWD station observations of wind, air temperature, relative humidity, and precipitation can provide a quality-controlled regional context and support gap and sensitivity analyses, but should not be treated as measurements of the air approaching the barn after modification by nearby buildings, vegetation, and terrain. CFD could then be evaluated against the empirical spatial patterns and used to examine unmeasured airflow paths, but such modelling would constitute a new validation study rather than evidence produced here. Finally, emission applications require simultaneous assessment of representative airflow and concentration at an appropriate control surface; the present concentration ranking alone is insufficient for that purpose.

5 Conclusions

Concentrations in the investigated naturally ventilated dairy barn remained heterogeneous across both height and horizontal position, and the persistence of that structure was gas-specific. Temporal aggregation reduced short-term location error but did not remove the comparatively strong spatial component of ammonia and its ratios. A measurement period should therefore not be lengthened as a substitute for spatial coverage without first demonstrating that the remaining location bias is acceptable for the intended gas and outcome.

The dense campaign showed that a distributed top-and-bottom configuration could preserve the internal CO_2 , CH_4 , and NH_3 reference means while retaining vertical contrast and reducing sampling effort. Dense initial mapping followed by explicit, response-specific configuration testing is therefore supported as a practical design strategy. A single physically screened location could provide a useful site-specific approximation, but it could

not replace spatial averaging or be assumed representative of emission flux. Independent seasonal validation with concurrent airflow and operational measurements remains necessary before the reduced configuration or single-point candidate is transferred beyond this barn.

Acknowledgements

The technical and animal-care staff of the experimental dairy facility are gratefully acknowledged. *[Funding project names, grant numbers, and specific technical contributors are to be confirmed before submission.]*

Author contributions

Harsh Sahu: conceptualisation, methodology, investigation, data curation, formal analysis, visualisation, and writing—original draft. Long Chen: methodology, formal analysis, and writing—review and editing. Thomas Amon: conceptualisation, supervision, funding acquisition, and writing—review and editing. Tariq Mehmood: investigation and writing—review and editing. Aditya Rawat: investigation and writing—review and editing. David Janke: conceptualisation, methodology, supervision, project administration, and writing—review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The processed data and analysis code will be made available in a public repository upon acceptance. *[Repository and DOI to be added.]*

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